

Designing Access in Sound Art Exhibitions: Centering Deaf Experiences in *Musical Thinking*

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ABSTRACT

This case study details the process of generating recommendations for the implementation of specific haptic technologies in an upcoming exhibition of sound and video art in order to improve the museum experience for Deaf and Hard of Hearing patrons. Various haptic technologies and intensity settings were evaluated by D/deaf and Hard of Hearing participants through a combination of structured user experience surveying and a focus group. Insights gained from this mixed-methods approach were then used to generate recommendations for specific vibro-tactile technologies for each artwork in the exhibition. Additionally, general design insights into designing more accessible sound-art experiences, such as the need for tailored haptic signal design instead of using native audio signals, were also provided.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in accessibility; Accessibility design and evaluation methods; • **Applied computing** → Sound and music computing.

KEYWORDS

Accessibility, Haptics, Sound Art

ACM Reference Format:

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1 INTRODUCTION & BACKGROUND

This case study generates design insights on how technology may be used to improve museum patrons' engagement with sound and video art. Specifically, we evaluate technologies that may improve

the accessibility and overall experience of Deaf and Hard of Hearing (HoH) visitors to the Smithsonian American Art Museum's (SAAM) upcoming exhibition entitled *Musical Thinking: New Video Art and Sonic Strategies*. This research was undertaken at SAAM's request, with goals developed in tandem with the Musical Thinking team, and with support from the Smithsonian Accessibility Innovation Fund. Based on the results of this mixed-methods study, evidence-based recommendations regarding accessible technology augmentations were made to the museum's curation team in preparation for this specific exhibition. Additionally, some generalized insights into designing more accessible sound-art experiences were also generated. Given the exhibit's exploration of music and sound, the use of this project's recommended technologies as well as the work of Deaf sound artist Christine Sun Kim in the exhibition highlights the importance of including Deaf perspectives and insights in this work. The results of this project will not be universal solutions or technocratic cures. At a minimum they will serve as devices that improve some exhibition-goers experience and, ideally, they will serve as a physical acknowledgment of people excluded from sound-focused spaces and as an invitation to consider the nuances of what can be done to make museums, galleries, and the artwork genuinely more inclusive.

Several previous exhibitions and technologies have highlighted Deaf perspectives. The recent exhibition "Activating Captions" from Argos Arts, curated by Christine Sun Kim and Niels Van Tomme, highlights the poetics, promise, and challenge inherent in assistive and augmentative technologies such as closed captions [2]. Artist Wendy Jacob and architect Hansel Bauman collaborated on a series of workshops and works with Deaf students from Gallaudet University and students from MIT's Autism Studio. A notable outcome of this collaboration was the creation of a "dance floor for a silent dance party" which involved a large-scale installation of various sub-woofers and a vibrating floor to establish a shared tactile experience of sound [6, 10]. Work has also been done to improve traditional museum experiences for HoH visitors through the evaluation and implementation of hearing assistive technologies such as training of museum volunteers and docents to improve speech clarity and additional amplification of their voice during a museum tour [11]. Equally important work has highlighted issues facing museum and gallery goers with sensory Disabilities, such as lack of accessible ways to engage with the work, and found that staff training methods that included members of the Disabled community

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aided in creating museum experiences that were more enjoyable for people with Disabilities [1, 9].

People who are D/deaf and HoH have unique relationships with sound that are typically not considered in the design of musical or other sound-focused experiences. A common method to improve the experience of D/deaf and HoH folks is to include elements that can be perceived through touch, such as vibro-tactile haptic devices and sound systems with large sub-woofers. Systems and strategies to augment user experiences with additional haptic information and feedback have been developed in the past several decades. The practice of holding balloons during concerts to provide richer vibrotactile feedback is not uncommon in the Deaf community [3]. The first major consumer exposure to vibro-tactile technologies that transduced vibrations directly onto user's skin was in 1997, when *Nintendo* released the *Rumble Pak* for the N64 controller [4]. Since then, vibro-tactile technologies have greatly evolved. The interactive fashion brand *CuteCircuit* developed the *SoundShirt*¹ and *HugShirt*² which use up to 28 haptic actuators sewn into a lightweight fabric to create a shirt that communicates vibro-tactile haptic information to the wearer. A variety of wearable vests and slings have been designed primarily for immersive gaming experiences, or as an augmentation to music listening. The vests include commercial projects such as the *Woojer Vest*³ and *Strap*⁴, the *BHaptics Tactsuit*⁵, and the *Subpac*⁶.

1.1 Theoretical Framework

While some members of the Deaf community may not personally identify as Disabled and have complex, nuanced relationships with Disability, the literature on Design for Disability is useful in establishing a theoretical foundation for this project [8]. Design for Disability centers on the non-normative experience of a Deaf and/or Disabled user and seeks to go beyond legal compliance to create an experience that actively provides opportunities for nuanced exploration of pride and other aspects of the Deaf and/or Disabled identity, as well as incite discussions among all stakeholders [5]. This framework guides designers to be co-creators alongside their end user and acknowledges that this work is not solely about pragmatic, physical access. The aesthetics, poetics, and politics of access technologies themselves are extremely important to consider when designing these technologies. For instance, consider the stark contrast between a narrow concrete ramp leading into a side entrance of a room versus the bold, bright orange, multi-story ramp that forms the centerpiece of the Ed Roberts building in Berkeley, California [7]. The first ramp's strongest aesthetic and political association is that of compliance, a pragmatic way to provide physical access and avoid litigation. The second ramp illustrates the architectural focus of the space is its relation to physical Disability. Disability Design reminds designers to resist the "universal" or "silver bullet" solutions that simplifies the rich, varied space of Disability access to a single technical solution. This framework reinforces the notion

that access cannot be "solved" as it is, and will always be an ongoing effort that evolves within communities and individuals.

Keeping this framework in mind, it is important to acknowledge that our mission with this work is to provide accessible technology solutions to improve the experiences of artworks that already exist, not to redesign the original materials in any way. The technologies recommended in this project are being incorporated as an augmentation to "remedy" the artwork/artists' inaccessible design choices. This is not to shame artists, who may be following industry conventions of, for example, not including open captions in their video works, or who may not have the knowledge or resources to deeply consider these issues. This acknowledgment is made to highlight that this project of augmenting and retrofitting is not, and cannot, be an endpoint. Access is something that more people should be actively thinking about at every stage of creating and displaying culturally significant works. The current reliance on imperfect, retrofitted solutions reinforces the harmful notion that access is an afterthought. This project is an encouraging step toward more stakeholders considering access more deeply and earlier on. This collaboration between two large, influential institutions will hopefully serve as an invitation to other institutions to start doing the same or to deepen their existing efforts.

2 METHODS

This project used a mixed-methods approach, combining a structured user-experience (UX) survey with an exploration- and discussion-based focus group. The scope of solutions being evaluated was narrowed by the research team and consulting partners to allow for this mixed-methods approach to generate specific, applicable insights while still maintaining some generalizability. However, certain implementation-focused aspects of the solution space, such as the placement of haptic drivers, were not as tightly constrained and would benefit from a more exploratory and group-discuss approach. Additionally, the focus group allowed for discussion and meaningful inclusion of additional members of the Deaf community into the process. Sessions were facilitated in ASL or spoken English, dependent on participant preference. ASL interpreters were present at all group sessions to ensure access to discussions between all participants. All methods were approved by the Internal Review Board of Gallaudet University.

2.1 Partnership details

This project was a collaboration between the Motion Light Lab (ML2) at Gallaudet University and the Smithsonian American Art Museum (SAAM), with additional team members from Stanford University. Initial planning, brainstorming, and technology short-lists were worked on jointly with the entire team. The user evaluation and resultant recommendations were carried out largely by ML2 staff and a team member from Stanford University.

2.2 Evaluation Environment

Both the User Experience (UX) survey and focus group portions of the study were conducted in the Motion Light Lab at Gallaudet University. The testing room and equipment were selected and prepared to mimic the video display rooms in SAAM. This was done to ease the translation into the physical gallery space. The

¹<https://cutecircuit.com/soundshirt/>

²<https://cutecircuit.com/hugshirt/>

³<https://www.woojer.com/pages/vest>

⁴<https://www.woojer.com/pages/strap>

⁵<https://www.bhaptics.com/tactsuit>

⁶<https://subpac.com/>

testing room was darkened to be below 1 Lux when the projector was off, measured from the seated user position 3.5 m from the screen. An Epson Home Cinema 1060 (3100 Lumens) projector, similar to the models used by SAAM, was used to project all videos onto a medium-gray polyester backing screen. These distances and materials were chosen to approximate the conditions one might anticipate in the galleries where the various artworks would be exhibited. A 5' x 2' x 1' metal-framed sofa with 3" thick seat cushions and black faux-leather cover, and a 4' x 2.5' x 1' wooden bench with a backrest, armrests, and 2" thick cushion were used as seating options during the study. The audio was played by a pair of JBL stereo speakers and a Polk Audio PSW 10" sub-woofer, set to a play-back volume of approximately 85 dB SPL.

2.3 Technologies Evaluated

Technologies were selected after several rounds of art review and brainstorming meetings between Gallaudet University and SAAM. Multiple technologies were short-listed and evaluated based on their safety for public use, cost, and potential aesthetic-fit with the exhibition artworks. Notable exclusions included the Lasercube-brand laser projection (excluded based on safety concerns), Ultraleap-brand mid-air haptics (excluded based on cost and aesthetic concerns), as well as air-cannon-based haptic solutions (excluded based on aesthetic-fit, feasibility, and safety concerns). Creative captioning technologies, such as captions being displayed in non-traditional regions of an exhibition with varying colors and fonts, were considered. Ultimately these solutions were excluded due to time constraints and feasibility concerns.

While the initial short-list of technologies included a variety of visual and vibro-tactile technologies, the final list of technologies was limited to four vibro-tactile technologies: (1) Wrist haptics: The Neosensory Buzz vibro-tactile wrist band, (2) Haptic seating: a sofa or bench with a vibro-tactile transducer secured to it, (3) Haptic floor panel: A 12" x 18" x 6.5" wooden box with vibro-tactile transducers attached to the 0.5" thick acrylic top panel of the box, and (4) Passive haptics: Standard balloons and inflatable balls of four dimensions (18", 12", 10", and 5" diameter) that passively resonate in response to acoustic pressure waves (i.e. sound).

The vibro-tactile transducers evaluated in the haptic seating included the ButtKicker Advance BK4-4 transducer Dayton Audio BST-1, BST-2, and TT25-8 (Dayton Audio, Springboro, OH, USA). The same transducers were evaluated in the haptic floor panel configuration. All active haptic devices were sent signals amplified by a ButtKicker Power Amplifier BKA300-P amplifier.

2.4 Participants

Of the ten participants recruited for the UX survey study, seven self-identified as d/Deaf, two as HoH, and one as hearing. Three participants used hearing assistive technologies, and the mean age of the participants was 23 ± 1.2 years.

Nine participants were recruited for the focus group, with six participants from Gallaudet University self-identifying as d/Deaf, and three participants self-identifying as hearing, two from the SAAM and one from Gallaudet University. The focus group consisted of people familiar with the project and the technologies involved from both Gallaudet University and SAAM. This decision

was made to allow the focus group to view the actual artworks, allowing for increased nuance and details in the group discussions and explorations.

2.5 UX Survey

Three devices were selected for evaluation in the UX survey portion of the study: the wrist haptics, haptic seating, and haptic floor panel. Passive haptics were excluded from this portion of the study due to the lack of clear consensus on the exact material and dimensions of the passive haptic objects. Time constraints also prevented each participant from using each of the four different passive haptic objects in every genre. As the nine artworks that will be included in the exhibition could not be shown to a wider audience prior to the exhibit, mock videos were created to be roughly representative of the artwork by sourcing videos from the same genre. The seven genres identified to be emblematic of the artworks contained in the exhibition were: Latin Percussion, Rhythmic Gunfire, Opera, Ambient Natural Field Recordings, Jazz Piano, Contemporary Hip-Hop, and Instrumental Disco. Multiple videos from each genre were sourced from public online repositories, and the researchers selected the videos that most closely resembled the artwork. Three mock videos were created from the selected videos: a baseline video containing 10-15 second excerpts from each genre, a main video, containing 30-45 second excerpts from each genre except Instrumental Disco, and a disco video containing a 45-second excerpt of Instrumental Disco. The Disco video was separated from the main video to allow sufficient time for participants to comfortably transition to the haptic floor panel from the haptic seating. Additionally, a title screen was placed between each genre clip to provide participants with context and for ease of survey completion. Participants used the haptic wristband and haptic seating during the main video, and the haptic wristband and haptic floor panel during the disco video. This distinction was made due to constraints in the final exhibition space limiting which technologies could feasibly be placed in each exhibition. The haptic wristband was evaluated in every condition as it was deemed infeasible to have a user take the device on and off for individual artworks.

Participants first watched the baseline video and were then randomly sorted into two balanced groups (A or B). Participants were always shown the main video twice, with different haptic intensity levels on the devices used, and then the disco video was shown twice, again with differing haptic intensity levels. Group A experienced high-intensity wrist haptics and low-intensity haptic seating or floor panel first, while group B experienced the opposite order. The low haptic intensity level was determined by the research team to be the lowest setting where the vibrations were clearly perceived by all members of the team. The high haptic intensity level was set to be the maximum level the technology could safely output without experiencing hysteresis or perceptible distortion artifacts.

At the end of the baseline video as well as the end of each genre clip in the main and disco videos, participants were asked to write down any overall comments they had and answer the following questions on a five-point Likert scale ranging from strongly disagree to strongly agree for each technology used in each clip:

- Improved my overall experience
- Helped communicate emotion or mood

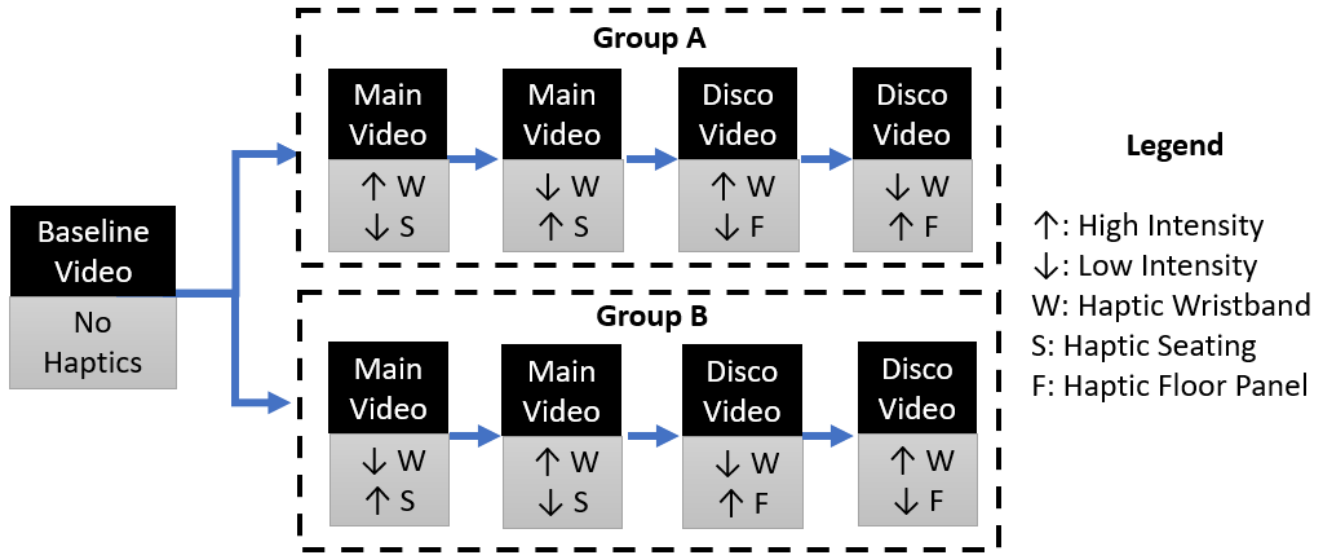


Figure 1: UX survey experiment flow.

- Was overwhelming or distracting

At the end of the UX survey, participants completed a short exit survey consisting of a 5-point Likert response to “Were you able to interpret any emotion in the music that was not relayed to you via sound?” as well as a series of structured, open response questions asking participants overall thoughts about the individual technologies and their thoughts on the effect of genre on their experience of the technologies.

2.6 Focus Group

Participants all arrived at the same time and collectively participated in each phase of the focus group. Participants’ reactions, comments, and questions were noted by the research team and discussion was encouraged among participants. The focus group structure consisted of three phases, namely:

(1) The *Clubbing* exploration phase: Gutierrez’s *Clubbing* was played on repeat as participants were encouraged to stand on the haptic floor panel and alter the panel’s intensity setting as well as grab any number of the passive haptic balloons/balls and explore the space.

(2) The *Adal* haptic wristband phase: Participants were asked to wear the haptic wristband while the *Adal West Side Story Upside Down, Backwards, Sideways and Out of Focus* piece was projected. The audio was played at a lower volume and high-pass-filtered to approximate the sound that would come from the artwork’s small, embedded speakers. Additionally, a clip of ambient museum background noise was played in the background to simulate the other sounds present in the space that the wristband would detect and react to. During the focus group, members of the SAAM team noted that the choice of background was not ideal as it contained too much HVAC noise and was overall louder than what is typical of the space.

(3) The haptic seating driver position evaluation phase: Gutierrez’s *Clubbing* was again played on loop while participants were asked to sit in pairs on either the sofa or wooden bench while the Dayton Audio BST-1 vibro-tactile transducer was positioned in various locations, including the backrest, bottom support beams, and bottom panel.

3 RESULTS

We present insights from all facets of the study on each of the four technologies evaluated in this section. It should be noted that due to the small sample size, the differences seen the UX survey results table are not statistically different from one another. However, when paired with participant comments and focus group insights, they can effectively serve as an additional point of reference.

The wristband’s differing intensity levels appeared to elicit no clear preference from participants, nor any clear impact on their UX survey ratings. Many comments noted themes of disconnect or confusion, possibly also seen in the survey ratings as the effect of the genre seemed to have little effect on participant ratings. Usability and feasibility concerns were raised by members of the SAAM team during the focus group sessions and in meetings, such as the distribution and collection procedures for small, expensive devices proving potentially troublesome.

3.1 Haptic Seating

Participants expressed a clear preference for higher intensity settings seen both in the UX survey results, comments, and in the focus group. Additionally, genre effects were clear. Highly rhythmic genres with significant low-frequency content such as Hip-Hop, Gunshots, Jazz, & Latin Percussion were highly rated, while Opera, with no low-frequency content, was rated mildly and Nature, with very little rhythmic content, was rated low. In the focus group, the

	Wristband - HIGH			Wristband - LOW		
	Improve Overall	Communicate Emotions	Overwhelm	Improve Overall	Communicate Emotions	Overwhelm
Latin Perc	3.3	3.3	2.3	3.7	3.4	1.9
Gunshots	3.8	3.8	2.4	3.7	3.3	2
Opera	3.9	3.8	1.8	3.6	3.7	1.6
Nature	3.9	3.7	2.1	3.4	3.4	1.6
Jazz	3.9	3.8	2.1	3.3	3.2	1.6
Hip Hop	3.8	3.8	2.2	3.6	3.6	1.9
Disco	3.6	3.6	1.9	4.4	4.2	2.1
Mean	3.74	3.69	2.11	3.67	3.54	1.81

Table 1: Mean scores of agreement with statements that the technology “improved overall experience”, “helped communicate emotion or mood”, and “was overwhelming or distracting”. Scores are presented by genre for the haptic wristband in both high- and low-intensity modes, where 5 is strong agreement and 1 is strong disagreement.

Session	Participant Number	Genre	Comment
UX Survey	UX-P7	Latin Percussion	“Feels annoying”
UX Survey	UX-P6	Opera	“Wristband was awesome”
UX Survey	UX-P5	Nature	Wristband appeared to be overwhelming or confusing (researcher note)
UX Survey	UX-P6	Latin Percussion	“Wristband took away from couch”
UX Survey	UX-P5	Nature	“Wristband isn’t synced up”
UX Survey	UX-P5	Nature	“Feels constant, like an air conditioner noise”
UX Survey	UX-P7	Overall Comments	“The wristband is a no-no”
Focus Group	FG-P2 & FG-P8	Overall	Observed difficulty putting the wristband on/off, assistance requested (researcher note)
Focus Group	FG-P1	Adal	“Feel off-balance with it only being on one wrist”
Focus Group	FG-P2	Adal	“Difficult to distinguish between someone talking and music”
Focus Group	FG-P3	Adal	“Overwhelming, needs a sensitivity warning”
Focus Group	FG-P5	Adal	“It’s nice that you can move around with it on”

Table 2: Notable comments and observations regarding the haptic wristband.

	Seating - HIGH			Seating - LOW		
	Improve Overall	Communicate Emotions	Overwhelm	Improve Overall	Communicate Emotions	Overwhelm
Latin Perc	4.8	4.8	1.3	3.3	3.2	2.2
Gunshots	4.6	4.6	1.8	3.1	3	2.3
Opera	3.6	3.6	1.6	2.6	2.5	1.7
Nature	2.2	2.2	1.6	2.9	2.8	1.8
Jazz	4.8	4.7	2.00	3.9	3.9	1.6
Hip Hop	4.7	4.6	2.2	4.0	4.0	1.9
Mean	4.12	4.08	1.75	3.30	3.23	1.92

Table 3: Mean scores of agreement with statements that the technology “improved overall experience”, “helped communicate emotion or mood” and “was overwhelming or distracting”. Scores are presented by genre for the haptic seating in both high and low-intensity modes, where 5 is strong agreement and 1 is strong disagreement.

wooden bench was perceived as facilitating a more intense experience of the vibration when compared to the sofa. The haptic driver placement, whether on the back rest, support frames on the bottom

of the seating, or on the base board, seemed to make little to no difference in the user experience besides from some small differences in the amount of vibration felt in the armrests that were noted twice

Session	Participant Number	Genre/Artwork	Comment
UX Survey	UX-P1	Latin Percussion	"I really feel it"
UX Survey	UX-P1	Hip-Hop	"I felt everything. Felt all the pain in the video"
UX Survey	UX-P1	Opera	"Can't feel the voice. I personally can't feel her emotions"
UX Survey	UX-P6	Latin Percussion	"I prefer the seating be [more intense] than the wristband"
UX Survey	UX-P6	Hip-Hop	"Makes me feel like a deaf child at church"
UX Survey	UX-P8	Hip-Hop	"Both wristband and couch make me feel weird. I felt nothing that made [sense to me as] music"
Focus Group	FG-P1-5, & FG-P7-9	Disco	Agreed with the statement, "The wooden bench has a more intense vibration"
Focus Group	FG-P2	Disco	"I feel a part of it. The sofa is awesome"

Table 4: Notable comments and observations regarding the haptic seating.

	Floor - HIGH			Floor - LOW		
	Improve Overall	Communicate Emotions	Overwhelm	Improve Overall	Communicate Emotions	Overwhelm
Disco	5	5	1.7	4.1	3.6	2

Table 5: Mean scores of agreement with statements that the haptic floor "improved overall experience", "helped communicate emotion or mood" and "was overwhelming or distracting". Scores are presented by for the instrumental disco for the haptic floor panel in both high and low intensity modes, where 5 is strong agreement and 1 is strong disagreement.

Session	Participant Number	Genre	Comment
UX Survey	UX-P1	Disco	"I love it"
UX Survey	UX-P2	Disco	"[The] floor [is] good at distinguishing sounds but [it] is also slightly ticklish"
UX Survey	UX-P6	Disco	"I feel like I'm at a bar or a club"
UX Survey	UX-P6	Disco	"MUCH BETTER!!" (When intensity was raised)
Focus Group	FG-P3	Disco	"The haptic floor is my favorite"
Focus Group	FG-P7	Disco	"I wanna dance"
Set-Up	Member of Research Team	Disco	"I think my shoes [dampen] the vibrations" (Note: participant was wearing cork platform heels)

Table 6: Notable comments and observations regarding the haptic floor panel.

with no apparent pattern. The Butt kicker Advance BK4-4 haptic driver, which was attached to the back left leg of the sofa, appeared to facilitate a slightly more intense experience of the vibrations when compared to the Dayton Audio BST-1. However, the Dayton Audio drivers were not strongly coupled to the seating as they were being physically held in place by a research team member. Additionally, we noted that placing the driver on the wooden base board led to more audible sound transduced into the air through acoustic bleed.

3.2 Haptic Floor Panel

The haptic floor panel on a high-intensity setting was unanimously rated as improving every participant's experience in the UX survey portion of the study. Additionally, it was noted in comments and in focus group testing that the haptic floor panel was "immersive", "moving", and some participants' "favorite" (UX-P3, UX-P5, and FG-P3). Some slight, subtle perceptual effects were noted depending on where participants stood on the panel, likely due to the participant's weight distribution on the panel affecting its dampening profile. Two participants noted that the vibrations on their feet felt ticklish

at times. The type of shoe and thickness of the sole appeared to impact the intensity of the vibration perception.

3.3 Passive Haptic Objects

Focus group participants' responses to exploring the space with the various-sized balloons and beach balls showed mildly positive reactions. As participants explored the space with the passive haptic objects there was a clear preference to be closer to the speaker, as it seemed to elicit a more intense and clearly perceptible vibration in the passive haptic object. Participants generally held the passive haptic objects with both hands, occasionally opting to hold two small passive haptic objects simultaneously, one in each hand. There was also a clear preference for the balloons with a thinner membrane over the beachballs which has a thicker membrane that did not translate vibrations as well. Participants generally expressed a preference for the medium (12" and 10") balloons, followed by the smaller 5" balloons, and then the large 18" balloon. A few participants noted the clear beach ball with silver glitter inside "looked very disco" (FG-P4).

4 DISCUSSION

This case study aimed to identify pairings between artworks in an upcoming exhibition and haptic technology that would improve the experience for D/deaf and HoH patrons. This was evaluated through a mix of UX surveys and focus-group discussions. The haptic wristband technology seems an inappropriate fit for a museum showing multiple sound-art exhibits of differing style due to the lack of strong, direct interpretation of haptic signals as well as a weaker emotive or immersive experience reported, additionally presenting several logistical and practical challenges.

The haptic floor panel was unanimously the highest-rated technology and it was apparent through user reaction and feedback that the technology was very well-received. Possible reasons for this include the strong thematic link between a thumping dancefloor and the experience of a disco club. Additionally, the technology affords active participation via dance. The type of shoes participants wore impacted their experience, and we anticipate that some future users may be inclined to remove their shoes. Notably, this technology was not tested with wheelchair/powerchair users due to prototype constraints. While the final installation will be wheelchair accessible, due to the lack of wheelchair/powerchair user feedback, exploring additional mobile or height-adjustable haptic solutions is highly recommended.

Haptic seating, as previously noted, provided a rich user experience, particularly when paired with artworks containing strong rhythmic and low-frequency elements. Likely as a result of the choice to send the generic audio signal designed to be listened to with one's ears into the haptic system without significant pre-processing may be an inappropriate choice. These audio signals not designed for the unique frequency response of the various devices, they were also not designed to be perceived through the sense of touch. *Haptic Foley*, a phrase coined by the research team to describe the process of subjectively translating the raw audio signals into signals more appropriate for vibro-tactile perception, could prove extremely useful in improving user experience. Techniques such as filtering, pitch-shifting, and additional augmentative synthesis could be used to create a more easily interpretable experience. We note that conversion from one medium to another, or any augmentation of an artwork, is not a neutral act. This process, much like that of captioning and written language translation, is not an exact science but rather a site of both information loss and generation. Overall, the haptic seating received a positive response from participants and has an added benefit of being relatively easy to implement and monitor.

Overall, participants' response to the passive haptic objects was only slightly positive. The novelty of the active haptic technologies, such as the haptic wristband and floor panel, may have introduced a novelty bias against the more mundane objects, such as a balloon, used for passive haptics. However, the strong use case of extending haptic accessibility to those who are not serviced by the seating or dance floor serves as a compelling reason to consider additional haptic solutions in the future, such as passive haptics or height-adjustable vibro-tactile pads, that may serve communities such as wheelchair and powerchair users more effectively. Notably, all solutions recommended are opt-in solutions, in which

no visitor is forced into using the technology. Vibration-based solutions might be overwhelming or painful for those with sensory processing or physical disabilities. Therefore, we recommended that non-vibrating options be present and that vibrating objects are clearly and consistently marked.

5 FUTURE WORK & CONCLUSIONS

Future work in this project includes: (1) the iterative process of Haptic Foley, whereby the signals that are sent to each haptic driver are individually designed, in a similar fashion that a sound designer or Foley artist would do for sounds in a film or video game; (2) Exploratory work designing a touch-pad interface for select exhibits; (3) facilitating the creation of signed interpretations of select works, ASL tours, and ASL video interpretations of wall text.

The goal of this case study was to gain design insights on how technology might be used to improve museum patrons' engagement with the artwork. Specifically, this project examined pairing accessible technology solutions with individual artworks in a specific exhibition and gain insight into general accessible design considerations for sound-art exhibitions. Our goal was not to find a one-size-fits-all accessibility solution. Instead, we sought to recommend technical solutions that would improve the experience of d/Deaf and HoH visitors, and possibly hearing visitors, while considering physical, financial, and aesthetic constraints. Additionally, we considered all our discussions across this multi-year collaboration as opportunities to spark new thinking and raise further questions about access, Deafness, design, and Disability within art museum contexts. The findings show the importance of building this process into the extended timeline for exhibition development. Initial conversations about a range of possibilities turned into a set of proposals to be tested; artists and their representatives were made aware of accessibility goals and engaged in solution discussions as soon as exhibition layout and design began. User testing and evaluation was completed in time for recommendations to be incorporated into exhibition budgeting, equipment ordering and installation planning. Prioritizing an inclusive exhibition experience from the outset is key to allowing tailored solutions to be developed and implemented cohesively.

Hopefully, in the future, galleries and museums will have clearer accessibility processes, standards, and expectations that go beyond basic requirements and encourage creative, collaborative approaches. As accessibility considerations across the arts become more prominent, it is hoped that augmentative, technocratic solutions will not be as needed if artists are designing accessibility into their artworks and installation preferences from the beginning.

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A TABLE OF RECOMMENDATIONS

Artwork	Representative Genre	Recommendation	Confidence level
Gutierrez, <i>Clubbing</i>	Instrumental Disco	Haptic dancefloor. Additional passive haptic objects aimed at wheelchair/powerchair users	Very High
Chacon, <i>Report</i>	Rhythmic Gunshots	Haptic seating (Buttkicker if available)	High
Jaffa, <i>Love is the Message, The Message is Death</i>	Hip-Hop	Haptic seating (Buttkicker if available)	High
Leigh & Laser, <i>Breakdown</i>	Opera	Haptic seating, extensive Haptic Foley suggested	Medium
Smith, <i>Pilgrim</i>	Jazz Piano	Haptic seating, would benefit from Haptic Foley. Explore visual vernacular interpretation	Medium
Adal, <i>West Side Story Upside Down, Backwards, Sideways and Out of Focus</i>	Latin Percussion	Captions on a 45 deg surface near the suitcase, explore the concept of a "touch pad"	Medium
Smith, <i>Sojourner</i>	Multiple	Haptic seating, would benefit from Haptic Foley	Medium
Ghani & Kelly, <i>When the Spirits Moved Them, They Moved</i>	Nature	Haptic seating, extensive Haptic Foley suggested	Low-Medium

Table 7: Accessible technology recommendations for each individual artwork as well as the representative genre associated with the artwork and the confidence level with which the recommendation has been made.